

"Preferential" Trade Agreements?

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Abstract

Preferential Trade Agreements (PTAs) are generally understood to promote trade by granting member countries preferential tariffs relative to third countries. We show that this assumption can break down during implementation. When PTAs phase out tariffs gradually, unilateral most-favored nation (MFN) tariff reductions can undercut negotiated preferential rates. We find that half of the 356 PTAs notified to the WTO do not adjust preferential rates following unilateral MFN reductions, making such reversals potentially common. Using Colombian transaction-level data from the U.S.–Colombia PTA, we show that 90% of products subject to gradual phase-out had MFN rates below their original base rates even before the agreement entered into force, and that 35% of imports expected to benefit did not effectively receive a preference in the early years. A triple-difference analysis shows that affected U.S. imports underperformed unaffected U.S. imports by 34%, relative to imports from third countries. This effect persists even after preference margins are restored.

JEL: F1, F10, F13, F14

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Introduction

Preferential Trade Agreements (PTAs) are generally understood to promote trade among member countries by granting them preferential access to one another's markets (Baier and Bergstrand, 2007; Bagwell and Staiger, 2016). Since Viner (1950), the trade-creating or trade-diverting effects of PTAs have been understood to arise from the lower tariffs granted to partner countries relative to the most-favoured-nation (MFN) rates applied to third countries. We show that this assumption can and often does break down during agreement implementation: *PTAs often fail to generate effective tariff preferences.*

The reason is that preferential tariff schedules are usually negotiated based on the MFN tariffs prevailing at the time, which countries remain free to change unilaterally. In many PTAs, tariffs are phased out gradually from this initial MFN baseline rather than eliminated immediately. If a government subsequently *reduces* its MFN tariffs, the resulting MFN rate can fall below the preferential tariff specified by the PTA. Some PTAs prevent this by updating the preferential schedule following MFN reductions. Others, however, continue to phase out tariffs from the original baseline. In these agreements, the PTA temporarily provides no tariff advantage relative to third-country imports until scheduled preferential reductions catch up with the lower MFN rate and restore a positive preference margin. To the best of our knowledge, we are the first to systematically document this institutional design feature across PTAs and show how it differentially shapes effective tariff preferences during agreement implementation.

First, we examine how widespread these two approaches are. We collect data on the legal treatment of unilateral MFN reductions and the presence of tariff phase-outs for all 356 PTAs notified to the WTO as of March 2026. We show that 80% ($n = 283$) phase out tariffs over time and, among these, 61% ($n = 174$) do not provide for preferential tariffs to be updated following unilateral MFN reductions. Although the incidence of updating provisions varies substantially across regions, this variation is not systematically related to economic size, institutional capacity, the tariff environment, or the degree of economic integration. Moreover, while updating and non-updating PTAs were similarly prevalent until the late 2000s, newer PTAs increasingly do not contain such updating provisions.

Second, we use Colombian transaction-level import data between 2010 and 2022 and focus on the implementation of Colombia's PTA with the United States, implemented in May 2012. We find that 90% of baseline MFN rates indicated in the agreement's elimination schedule of products to be phased out over time were reduced even before the agreement had entered into force, and that 35% of Colombian

imports from the United States that were expected to receive preferential treatment did not, in practice, benefit from lower PTA tariffs in the early years of the agreement due to unilateral Colombian MFN reductions. We then implement a triple-difference (DDD) design to test whether the absence of effective tariff preferences affected imports from the U.S. during the implementation of the PTA. We focus on products for which unilateral MFN reductions resulted in zero or negative preference margins. The design compares the post-implementation change in U.S. imports of these products with that of products retaining positive preference margins, relative to the corresponding changes in imports from non-PTA partner countries. Following implementation, U.S. imports of products without an effective preference underperformed those retaining a positive preference by 34%, relative to the corresponding difference among non-U.S. suppliers. Moreover, the effects of the temporary absence of preferences persist even after preferential margins are restored. Our event-study estimates show that U.S. imports in affected products decline relative to products retaining positive preference margins during the period in which unilateral MFN reductions eliminate the tariff advantage. This relative decline persists even after the PTA's scheduled tariff reductions restore positive preference margins. The persistence suggests that a temporary absence of preferential access can have lasting effects on bilateral trade flows, beyond the period in which the tariff advantage is absent.

These findings show that the consequences of PTA design extend beyond the tariff schedules negotiated at the time of agreement formation. Whether agreements adjust preferential rates when MFN tariffs change determines not only whether firms receive the intended tariff advantage during implementation, but also whether the temporary loss of that advantage has lasting effects on bilateral trade.

Thus, we show that PTA tariffs, despite their name, do very often not grant “preferential” market access for importing firms in the member countries. The findings have important implications for our understanding of the economic gains from preferential trade agreements as well as the political economy of drivers of preferential trade liberalization. They also yield important insights for policymakers regarding the institutional design of future PTAs.

First, if analyses take negotiated elimination schedules for granted without taking into account the legal treatment of unilateral MFN reductions, they might either mismeasure whether a particular good enjoys *de facto* preferential treatment, or they might under-estimate the gains in trade by mixing up products that enjoy actual preferences with products that enjoy no preferences due to unilateral MFN cuts. Second, our findings speak to the long-standing debate over whether PTAs are “building blocks” or “stumbling blocks” for multilateral trade liberalization (Bhag-

wati, 1991). We highlight an important but overlooked institutional design feature of PTAs: the treatment of unilateral MFN reductions during agreement implementation. In the absence of updating provisions, MFN liberalization can eliminate the tariff advantage enjoyed by PTA partners, extending the same market access to non-members. Updating provisions, by contrast, preserve preferential margins from such multilateral liberalization by adjusting preferential tariffs alongside reductions in MFN rates. PTA design can therefore determine whether unilateral MFN liberalization translates into less discriminatory market access or whether the preferential margin created by the agreement is preserved.

The remainder of the paper proceeds as follows. First, we discuss the related literature on the dynamic effects of PTAs, preference erosion, and the interaction between multilateral and preferential trade liberalization. Second, we show conceptually how unilateral MFN changes affect preferential tariffs in PTAs under different institutional designs: static vs. updating tariff base rates. Third, we document the extent of the phenomenon using agreement-level and transaction-level import data, and its effects on trade flows under the U.S.-Colombia PTA. Finally, we discuss the implications of our results and conclude.

1 Related Literature

This paper relates to several strands of research on preferential trade agreements (PTAs) and tariff liberalization. We contribute to this literature by identifying an institutional feature of PTA design that is absent from existing empirical analyses: how agreements treat subsequent changes in most-favoured-nation (MFN) tariffs during implementation. This feature matters because the effective tariff preference created by a PTA depends not only on the negotiated tariff schedule, but also on the evolution of MFN tariffs relative to that schedule.

First, the paper builds on the classic literature on the trade effects of preferential agreements. Since Viner (1950), PTAs have been understood as potentially generating both trade creation and trade diversion by lowering barriers to imports from partner countries relative to those from nonmembers. The central object in this literature is the preferential tariff margin: the difference between the tariff applied to imports from a partner and the MFN tariff applied to imports from third countries. Subsequent empirical work documents substantial effects of PTAs on trade flows (Baier and Bergstrand, 2007). Our analysis highlights an important qualification to this framework. Even when the negotiated preferential tariff schedule remains unchanged, the preferential margin can disappear if a member subsequently reduces

its MFN tariff below the preferential rate. The relevant source of preferential market access is therefore not determined solely at the time an agreement is negotiated.

Second, the paper relates to the literature on the dynamic effects of PTA implementation and the gradual phase-out of tariffs. Baier and Bergstrand (2007) argue that the gradual phase-out of tariffs may explain PTAs' stronger effects on long-term trade elasticity, while more recent work explicitly studies how trade responds to the sequential implementation of negotiated tariff reductions (e.g., Besedes, Kohl and Lake (2020); Khan and Khederlarian (2021); Dong and Jestrab (2022)). This literature emphasizes that the timing and structure of tariff reductions are central to understanding the effects of PTAs. We complement this work by showing that the relevant tariff differential can change during implementation even when the negotiated schedule itself does not. In particular, subsequent MFN reductions can temporarily eliminate the preferential margin, so that products formally covered by a PTA need not receive preferential market access in practice.

Third, our argument connects to the literature on preference erosion. A large body of research studies how reductions in MFN tariffs can diminish the value of preferential market access, particularly for developing countries receiving unilateral preferences under schemes such as the Generalized System of Preferences (e.g., Alexandraki (2005); Francois, Hoekman and Manchin (2006); Hoekman, Will and Braga (2009); De Melo and Carrère (2010); Raimondi, Scoppola and Olper (2012)). We extend this literature in two ways. First, we show that preference erosion is not limited to unilateral preference schemes but can also arise within reciprocal PTAs. Second, the institutional treatment of MFN changes can determine whether such changes merely reduce the value of a negotiated preference or temporarily eliminate it altogether. This distinction is important because the tariff margin relevant for trade is endogenous to both the negotiated schedule and the agreement's rules governing subsequent MFN changes.

Finally, the paper relates to the literature on the interaction between preferential and multilateral tariff liberalization. Since Bhagwati (1991), a central question has been whether PTAs act as *building blocks* or *stumbling blocks* for multilateral trade liberalization. Theoretical work shows that the formation of PTAs can alter governments' incentives to set MFN tariffs and pursue subsequent multilateral liberalization (Richardson, 1993; Krishna, 1998; Bagwell and Staiger, 1999; Ornelas, 2005; Saggi, Woodland and Yildiz, 2013). Empirically, Limão (2006) and Karacaovali and Limão (2008) find that MFN tariff reductions are smaller on PTA-covered products in large markets such as the United States and the European Union, whereas Estevadeordal, Freund and Ornelas (2008) finds evidence of greater multilateral liberalization following PTA formation in Latin America. This literature therefore largely asks how

preferential liberalization affects subsequent MFN policy. We study the complementary institutional question: when MFN tariffs change after a PTA has been negotiated, how do the agreement's rules determine whether preferential tariffs continue to provide a tariff advantage?

2 How MFN Rate Changes Affect Preferential Rates

Although PTAs aim to liberalize tariffs on *substantially* all trade, they rarely eliminate all tariffs immediately upon entry into force. Instead, most agreements define a tariff-elimination schedule that specifies whether tariffs for a given product are eliminated immediately or phased out gradually. For each product, the schedule determines a base rate, which is typically the MFN rate applied to third-country goods at the time of negotiation, and a staging category that governs the phase-out timeline. Generally, phase-outs follow a linear trajectory, with tariffs reduced in equal increments over a specified number of years.

Absent changes in MFN rates during implementation, preferential margins, that is, the tariff advantage granted to goods originating within the PTA over third-country goods, remain strictly positive and increase monotonically over time. Figure 1 illustrates this benchmark case for a product with an initial tariff of 20% eliminated in ten equal steps.

However, MFN rates are not static. Governments retain the authority to *unilaterally* reduce MFN tariffs *after* a PTA has been concluded. When such reductions undercut the preferential tariff implied by the elimination schedule, the intended tariff advantage can disappear. PTAs differ in how they address this risk. Most agreements can be classified into two categories. The first links preferential tariffs *dynamically* to the prevailing MFN rate, updating the base rate, and thus the entire preferential schedule, in response to MFN changes. The second defines preferential tariffs relative to a *static* base rate fixed at the time of negotiation and independent of subsequent MFN reductions.

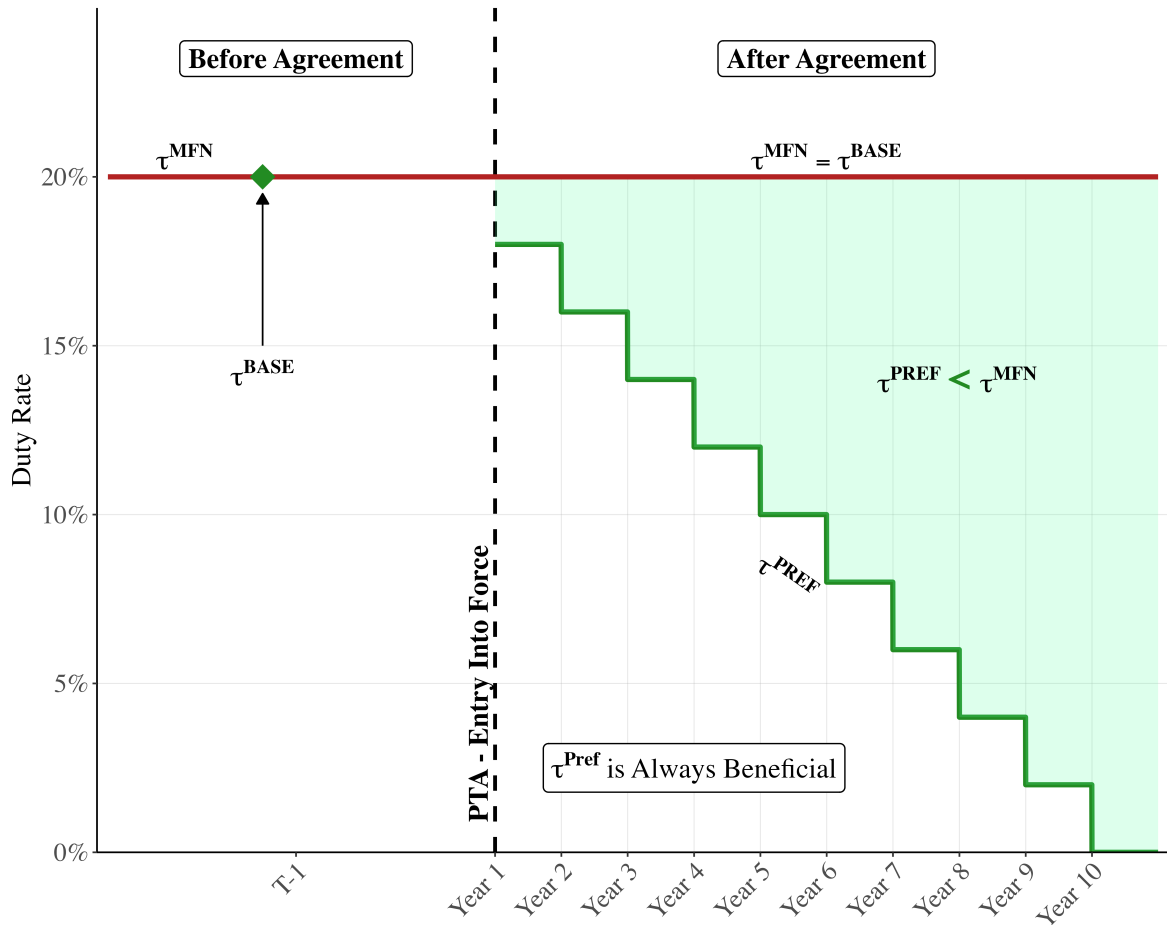


Figure 1
Gradual Tariff Elimination of a 20% Duty over 10 Years.

The elimination path is determined by two elements: the base rate, typically equal to the MFN rate at the time of negotiation, and the staging category, which specifies the phase-out period. In this benchmark case, the MFN rate remains unchanged, and the preferential margin increases linearly as the scheduled tariff reductions take effect.

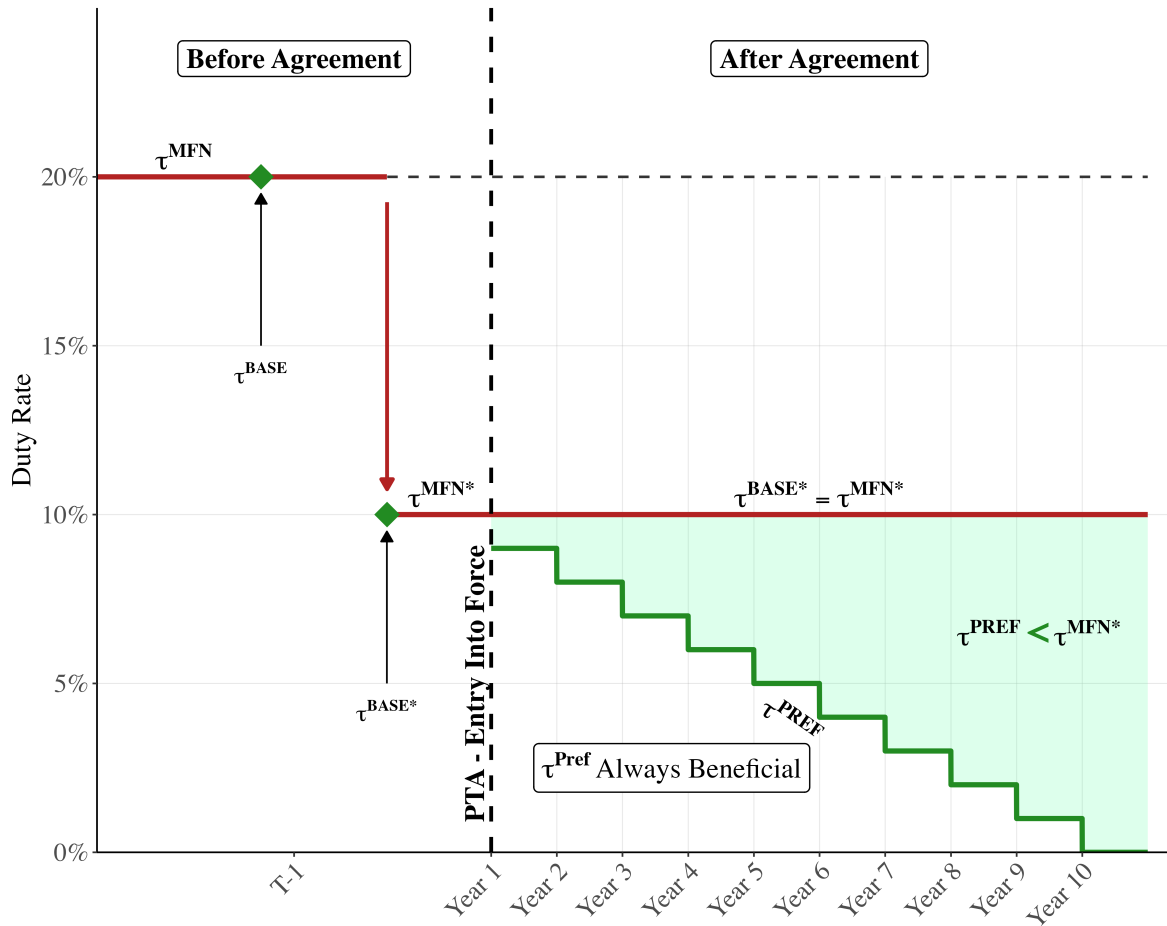


Figure 2

Tariff Elimination under MFN Changes - Updating Base Rate.

During the negotiations, the then-prevailing MFN rate, τ^{MFN} , is defined as the base rate, τ^{BASE} . At a later point, the MFN rate is reduced to τ^{MFN*} . The PTA stipulates that, in this case, the original base rate is updated to the new, lower MFN rate, τ^{BASE*} . Consequently, the annual preferential rate is calculated using the same preferential margin as originally intended, but now relative to τ^{BASE*} . In absolute terms, the annual tariff reductions are smaller; however, the relative tariff advantage over third countries is preserved throughout the implementation period.

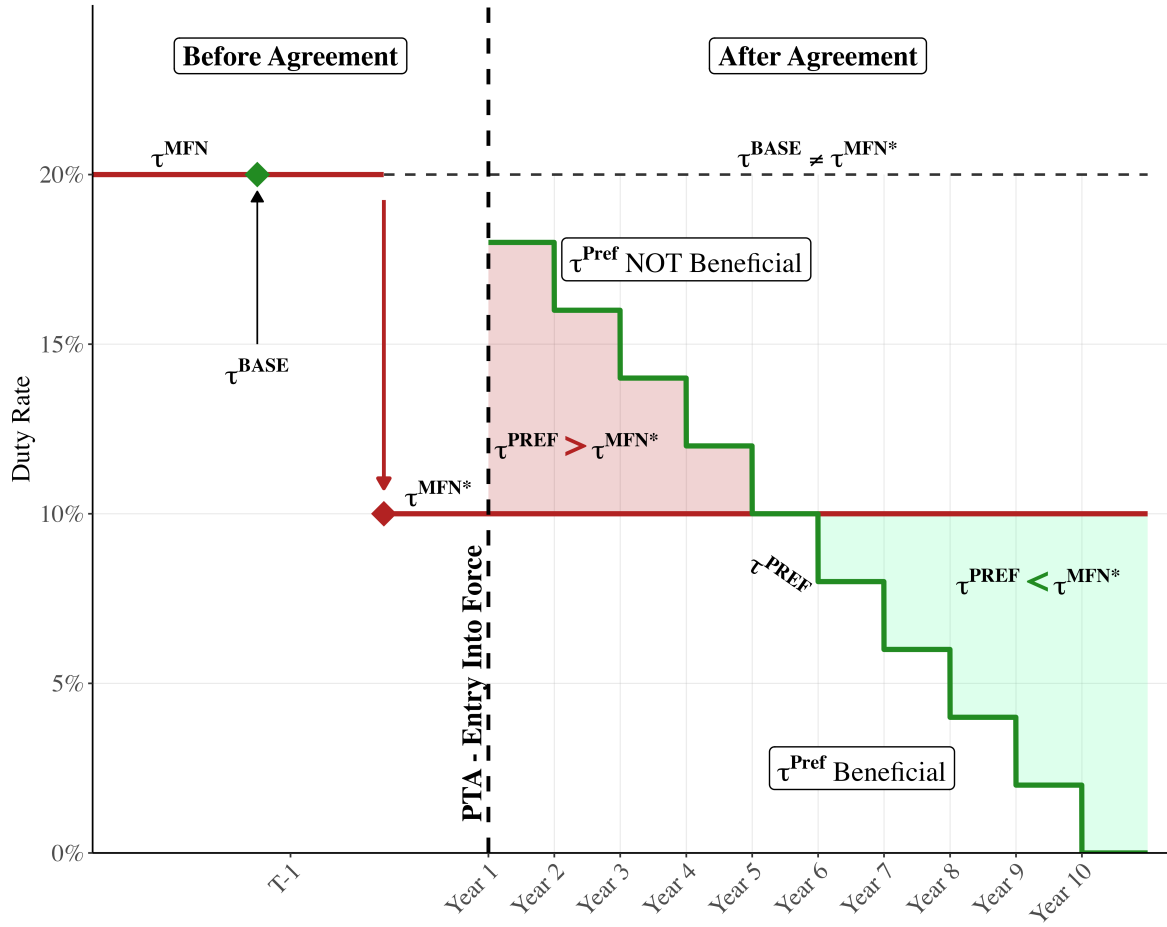


Figure 3

Tariff Elimination under MFN Changes - Static Base Rate.

During the negotiations, the then-prevailing MFN rate, τ^{MFN} , is defined as the base rate, τ^{BASE} . At a later point, the MFN rate is reduced to τ^{MFN*} . The PTA does not provide for an updating of the original base rate. Thus, the initial τ^{BASE} , on which the scheduled reductions are calculated, no longer corresponds to τ^{MFN*} . Consequently, the preferential rate, τ^{PREF} , exceeds τ^{MFN*} as long as the original elimination schedule has not yet caught up with the reduced MFN rate. In this case, the PTA confers no tariff advantage over third-country goods during the first six years of the agreement.

Whether a PTA adopts a dynamic or static base rate implies fundamentally different trajectories for effective tariff advantages. When base rates are updated, as in Figure 2, preferential tariffs remain below MFN rates and the negotiated preference margin is preserved. By contrast, under a static base rate, as in Figure 3, unilateral MFN reductions can undercut the preferential tariff. In such cases, the effective preference margin is eliminated, despite the agreement being formally in force. During these periods, a PTA ceases to confer any tariff advantage until scheduled reductions restore a positive margin. In economic terms, the agreement temporarily provides no discriminatory preferences relative to third-country imports, implying

that it need not generate additional trade in affected products. Despite its economic relevance, this institutional distinction between *static* and *updating* base rates has not been incorporated into the empirical trade literature. Existing studies rely on negotiated tariff schedules and typically assume a fixed MFN baseline during implementation (e.g., Besedes, Kohl and Lake (2020); Khan and Khederlarian (2021)). We document systematic cross-agreement differences in the treatment of MFN changes during phase-out periods and show that this feature has substantial implications for the measurement of effective tariff preferences and their trade effects.

3 Data

3.1 Agreement-Level Data

We construct a novel database covering 356 PTAs notified to the WTO and manually document (i) whether tariff liberalization is phased in gradually, and (ii) whether agreements update base rates and preferential rates following MFN reductions or instead rely on static base rates. The relevant provisions are typically located in a PTA's chapter on *Trade in Goods* or in the annex containing the tariff-elimination schedule. Table 1 illustrates the key difference in legal formulations underlying this classification:

Updating Base Rate	Static Base Rate
<i>If a Party reduces its applied most-favoured-nation customs duty rate ("MFN rate"), the schedule in Annex 2 of that Party shall apply to the reduced rates. If a Party lowers its applied MFN rate to a level below the base rate [...], that Party shall calculate the preferential applicable rate [...] on the lowered applied MFN rate, maintaining the relative margin of preference [...]</i> EU–Chile (Advanced Framework Agreement), Chapter 2, Article 2.5, Paragraph 3	<i>Where a Party reduces its most-favoured-nation applied rate of customs duty, that duty rate shall apply to an originating good of the other Party if, and for as long as, it is lower than the customs duty rate on the same good calculated in accordance with Annex 2-A.</i> EU–Japan (Economic Partnership Agreement), Chapter 2, Section B, Article 2.8, Paragraph 2

Table 1

Classification of PTAs in *Updating* vs. *Static* Base Rates

The table shows examples of how different PTAs treat the base rate, τ^{BASE} , when MFN rates are unilaterally adjusted. The left panel depicts the legal clause in the EU-Chile PTA, which foresees that the base rate is updated when MFN rates are changed. On the right, the EU-Japan PTA foresees no updating of τ^{BASE} .

The clause in the EU–Chile agreement stipulates that preferential rates are recalculated based on reduced MFN rates, thereby **maintaining** the relative margin of preference. In other words, the tariff-elimination schedule is updated in line with MFN reductions, ensuring that the originally negotiated tariff advantage is preserved. By contrast, the EU–Japan agreement provides that the reduced MFN rate applies as long as it remains below the preferential rate calculated from the original base rate. The base rate is therefore static, and when the MFN rate undercuts the preferential rate, the agreement ceases to confer any tariff advantage over third-country suppliers.

While formulations differ slightly across agreements, most can be classified as either *static* or *updating* base rate regimes. This distinction allows us to identify agreements in which unilateral MFN liberalization may (temporarily) eliminate negotiated preference margins. For agreements that lack a dedicated clause, we distinguish two cases. PTAs that do not include such a clause but specify a base rate in the tariff-elimination schedule are coded as *static*. By contrast, agreements that lack both a clause and explicit base rates, but instead define product-level margins of preference, are classified as *updating*. The final dataset will be made available for future research.

3.2 Colombian Transaction-Level Data

To assess the practical relevance of the described phenomenon, we use monthly transaction-level import data at the 10-digit national nomenclature level collected by Colombian customs (DIAN) for the period 2010–2022. Colombia concluded a PTA with the United States that entered into force on 15 May 2012. The agreement relies on *static* base rates and thus entails a risk of temporary preference elimination in the presence of unilateral MFN reductions.

Preferential rates for U.S.-originating imports are constructed using base rates and staging categories specified in Annex 2.3 of the U.S.–Colombia Trade Promotion Agreement. While the first phase-out step occurred upon entry into force, subsequent tariff reductions took effect on January 1 of each year. To account for potential intra-year changes in MFN rates, we compile a monthly series directly from the tariff portal of the Colombian customs authority.

Comparing preferential and contemporaneous MFN tariffs at the product–month level allows us to identify instances in which unilateral MFN reductions undercut preferential rates, thereby eliminating the effective preference margin. By constructing tariff-data from primary sources we mitigate concerns related to missing observations and aggregation in standard tariff datasets (Teti, 2024).

4 Results

In this section, we empirically investigate how unilateral MFN changes affect preferential rates granted in PTAs. First, we show to what extent legal formulations in all PTAs notified to the WTO foresee updating or static base rates (τ^{BASE}). Second, we use Colombian transaction-level data and the PTA with the U.S. as an example of an agreement with static base rates. We show how much trade is affected when MFN rates undercut preferential rates, resulting in negative preference margins, during the implementation phase of the PTA. Finally, we use a triple difference design to demonstrate to what extent trade gets diverted from PTA partners to imports from non-PTA countries in products affected by negative preference margins during the implementation of the PTAs.

4.1 Agreement-level

Figure 4 presents descriptive evidence on the prevalence of gradual tariff phase-outs and the different treatment of changes in MFN rates across PTAs. Gradual tariff phase-outs are a pervasive feature of PTAs, present in about 80% of all agreements (Panel A). Among agreements with gradual phase-outs, 61% define preferential tariffs relative to a static base rate and do not provide for updating in response to unilateral MFN reductions (Panel B). As a result, negotiated preference margins are not protected against subsequent MFN cuts in roughly half of all PTAs. Panel C further shows that this static design has become increasingly prevalent in recent agreements, including those still being phased in today. Taken together, these patterns indicate that most agreements phase out tariffs over time, while a substantial share rely on static base rates. Consequently in these cases, MFN reductions may eliminate preferential margins, so that the presence of a PTA does not automatically imply an effective tariff preference. Accounting for this distinction is therefore important in empirical analyses of PTAs' trade effects.

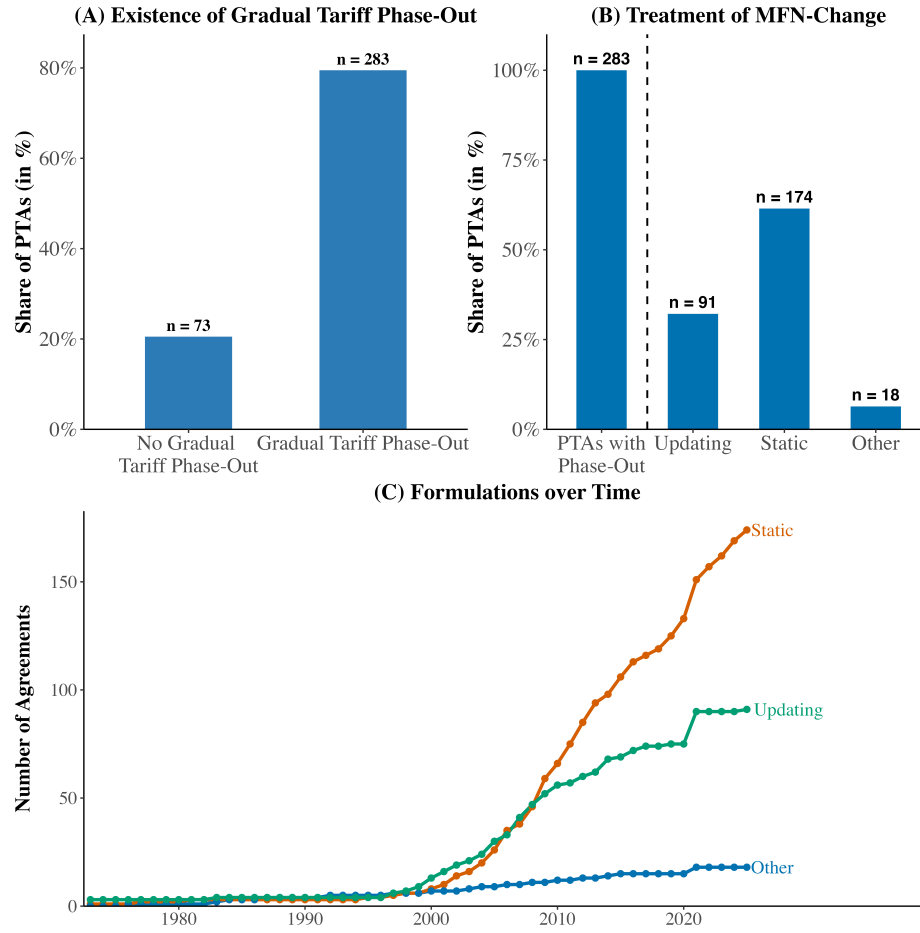


Figure 4

Treatment of MFN Reductions in Preferential Trade Agreements

Panel A shows that around 80% of all PTAs notified to the WTO contain some sort of gradual tariff phase-out. Of these agreements, around 60% do not update τ^{BASE} when MFN rates are adjusted (Panel B). Lastly, Panel C shows how the different treatment of MFN changes has developed across PTAs over time.

Table 2 shows a pronounced difference in MFN updating provisions across agreement configurations of developing and developed economies. MFN-updating provisions appear in 48% of South–South agreements, compared with 32.7% of North–South agreements and 14.3% of North–North agreements.

Table 2 MFN-Updating Provisions Across North–South Configurations

Configuration	Number of agreements Updating / Static	Updating (percent)	Examples (entry-into-force year)
South–South	36 / 39	48.0	<i>Updating:</i> CEFTA (2007); China–Pakistan (2007). <i>Static:</i> Pacific Alliance (2016); Peru–China (2010).
North–South	51 / 105	32.7	<i>Updating:</i> EU–Ukraine (2014); EFTA–Chile (2004). <i>Static:</i> US–Canada–Mexico (2020); US–Chile (2004).
North–North	6 / 36	14.3	<i>Updating:</i> EU–Chile Update (2025); GCC–Singapore (2013). <i>Static:</i> EU–Japan (2019); Korea–US (2012).

Notes: The table covers all 273 classified PTAs for which all variables are available. Percentages report the share of agreements within each configuration that contain an MFN-updating provision. North denotes World Bank high-income status in the year before negotiation; South combines all other income groups.

In Appendix 3, we examine whether the incidence of MFN-updating provisions is systematically related to observable characteristics of the negotiating parties and agreements. Although the prevalence of updating provisions varies substantially across regions, we find little evidence that this variation is explained by differences in economic size, institutional capacity, tariff conditions, negotiating complexity, or economic integration. Agreement-family, by contrast, accounts for substantial variation in the incidence of updating provisions. This points to regional and institutional drafting traditions as an important source of cross-agreement variation.

4.2 Transaction-level

We assess the economic relevance of this mechanism in practice by examining the Trade Promotion Agreement between Colombia and the United States. Negotiations began in 2004, concluded in 2006, and the agreement entered into force in May 2012. The PTA relies on static base rates and does not provide for dynamic updating following MFN reductions. The original base rates reported in the tariff elimination schedule correspond to Colombian MFN rates in effect on January 1, 2004. In a first step, Figure 5 plots the distribution of MFN tariffs at different stages of the PTA's negotiation and implementation for the three principal tariff-elimination categories (instant elimination, phase-out over 5 years, phase-out over 10 years). Together, these elimination categories account for 99% of tariff lines. The leftmost set of box-plots shows the base rates specified in the agreement, while the subsequent sets trace Colombia's four major MFN tariff revisions following the establishment of these base rates.

Already with the first MFN revision at the end of 2004, 5% of the base rates specified in the elimination schedule had been reduced on a multilateral basis. With the 2011 revision, Colombia significantly reduced its MFN tariffs across most products. Consequently, five months before the agreement entered into force, only 40% of tariff lines remained subject to the base rate originally specified in the elimination schedule. For products scheduled to be phased out over 5 or 10 years, around 90% of MFN rates were already below their respective PTA base rates. For these products, the tariff reductions negotiated under the PTA therefore did not translate into preferential market access upon entry into force; preferential treatment would only emerge once the scheduled PTA tariff phased out sufficiently over time and fell below the now prevailing, lower MFN rate.

Figure 6 illustrates the tariff-elimination schedule for U.S.-originating imports into Colombia as specified by the agreement based on pre-PTA trade-weighted import values. Around 60% of imports from the United States would have faced immediate tariff elimination upon entry into force in May 2012. For the remaining 40% of imports, tariffs are mostly phased out over 5, and 10 years. This share of trade would therefore be potentially exposed to the elimination of preferential margins through the unilateral MFN reductions by Colombia.

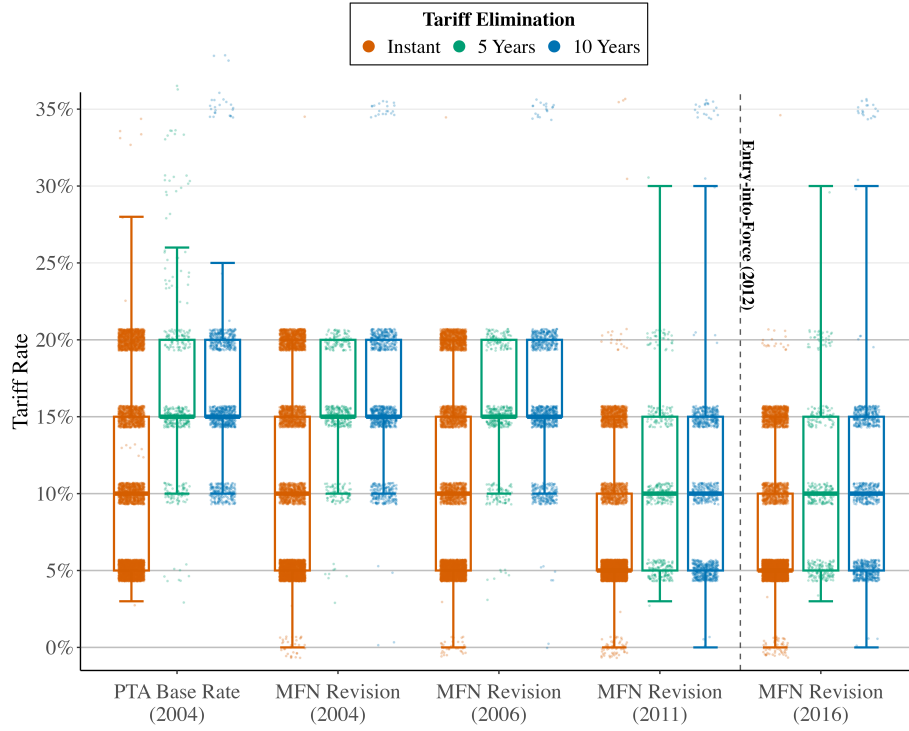


Figure 5

Evolution of Colombian MFN Tariffs Relative to PTA Base Rates

The figure illustrates the evolution of Colombian MFN tariffs relative to the base rates specified in the US–Colombia Trade Promotion Agreement, which correspond to MFN rates as of January 1, 2004 (leftmost group of boxplots). Subsequently, Colombia revised MFN rates in 2004, 2006, 2011, and 2016. Tariff lines are grouped by their negotiated elimination period in the US-Colombia agreement, with boxes showing the interquartile range and median and points representing individual tariff lines. The 2011 revision, preceding the agreement’s entry into force, substantially reduced MFN rates across all categories. By 2011, around 90% of tariff lines subject to gradual phase-out (over 5 or 10 years) already faced MFN rates *below* the base rates negotiated in 2004.

While Figure 6 describes the share of trade that was *potentially* exposed to the elimination of preferential margins, we now quantify how much trade was *actually* affected. The practical relevance is assessed by calculating the share of imports, across products i at time t , that received a tangible preferential tariff advantage, i.e., a positive preference margin:

$$\text{Tangible Preferential Share}_t = \frac{\sum_i \text{Imports}_{i,t} \mathbf{1}(\tau_{i,t}^{\text{PREF}} < \tau_{i,t}^{\text{MFN}})}{\sum_i \text{Imports}_{i,t} \mathbf{1}(\tau_{i,t}^{\text{MFN}} > 0)}. \quad (1)$$

The denominator is restricted to imports facing a strictly positive MFN tariff, as

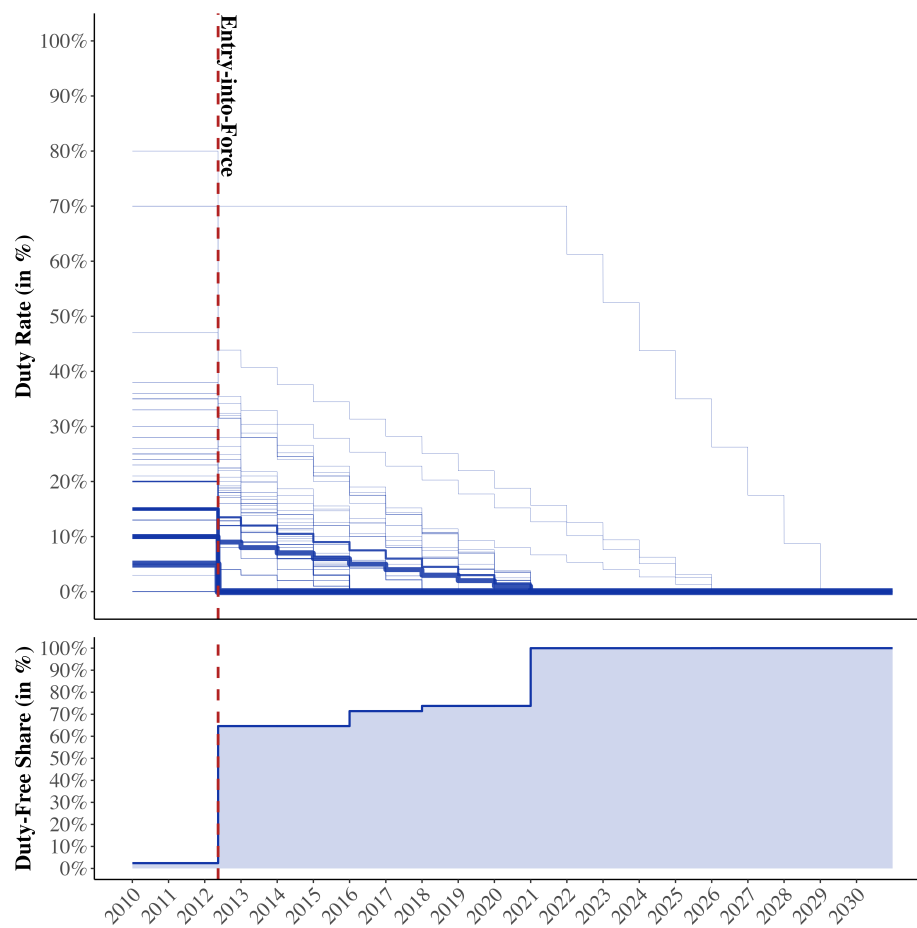


Figure 6
Colombia-US Elimination Schedule

The figure illustrates the tariff-elimination schedule for U.S.-originating imports into Colombia as defined in the US-Colombia Trade Promotion Agreement. Line thickness reflects the pre-PTA trade-weighted share of imports (January 2010–April 2012) across base rate–staging categories. Approximately 60% of pre-PTA imports are subject to immediate tariff elimination upon entry into force, while the remaining 40% are phased out over 5, 7, and 10 years. This latter share of trade is potentially exposed to the elimination of preferential margins through unilateral MFN reductions.

preferential margins can only arise when MFN tariffs are positive.

Figure 7 plots the monthly share of U.S.-originating imports into Colombia that receive a tangible preference after entry into force of the U.S.–Colombia PTA. As outlined in Section 2, unilateral MFN reductions by Colombia between negotiation and implementation lowered tariffs on third-country goods below the preferential rates implied by the agreement’s elimination schedule. As a result, negotiated preference margins for many products were effectively eliminated. Consequently, during the first three years of the PTA, between 20 and 35% of U.S.-originating imports facing positive MFN tariffs did not receive a tangible tariff advantage, despite being covered by the agreement.

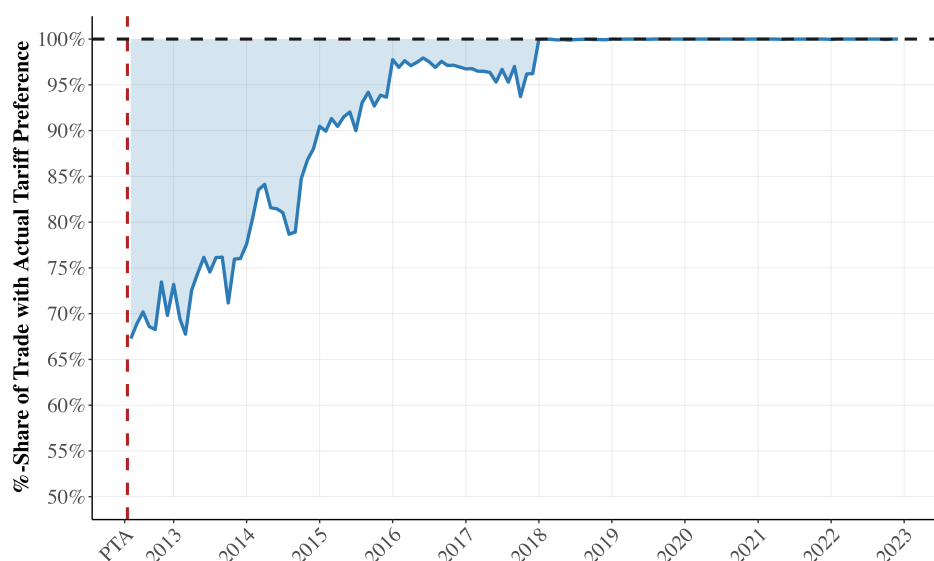


Figure 7

Effective Tariff Preferences under the U.S.–Colombia PTA

The figure plots the monthly share of U.S.-originating imports into Colombia that receive a positive preference margin following the entry into force of the U.S.–Colombia Trade Promotion Agreement. The share is calculated over imports facing strictly positive MFN tariffs. Due to unilateral MFN reductions by Colombia between negotiation and implementation, preferential rates implied by the agreement exceeded MFN rates for a substantial share of products and trade, eliminating effective tariff advantages. As implementation progressed and preferential rates declined below reduced MFN levels, preference margins were gradually restored.

Only as implementation progressed and preferential rates declined sufficiently to fall below prevailing MFN rates did negotiated “preferences” translate into effective tariff advantages. In practice, it took six years after entry into force for the PTA to deliver positive tariff advantages across all covered products. These patterns indicate that a substantial share of trade is affected by the elimination of effective preferences, underscoring the importance of accounting for this mechanism when assessing the

trade effects of PTAs.

4.3 Research Design

First, in order to estimate the effect of negative preference margins on diversion of trade from within the partnering PTA countries, we estimate the following linear model:

$$\log(Imports_{ict} + 1) = \beta (US_c \times Post_t \times NegMargin_i) + \alpha_{ic} + \gamma_{it} + \delta_{ct} + \varepsilon_{ict}. \quad (2)$$

where the outcome is logged imports in product i from country c in year t . US_c is an indicator equal to one if country c is the United States and zero otherwise and $Post_t$ is an indicator for the post-PTA period, when $t \geq 2012$. $NegMargin_i$ is an indicator equal to one for products that experience zero or negative preference margins at any point after the implementation of the US-Colombian PTA. The specification includes fixed effects for the product-country α_{ic} , and fixed effects for product-year γ_{it} and country-year, δ_{ct} .¹ As control units we use never-treated country-product pairs, that never have a PTA with Colombia. Thus, the coefficient β captures the differential change in U.S. imports after 2012 for products that experience zero or negative preference margins, relative to the corresponding change for products that retain a positive preference margin, using imports from non-PTA partner countries as the comparison group. We expect that Colombian imports from the U.S. will be no more attractive than imports from the rest of the world during agreement implementation, resulting in either a negative or non-significant β coefficient. Note that this differs sharply from standard assumptions that lower, negotiated preferential tariffs would boost imports from the U.S., compared to non-PTA partners, over time. (Besedes, Kohl and Lake, 2020; Baier and Bergstrand, 2007).

Second, we expect that the negative or null effect of MFN rates undercutting preferential rates should change over time as the agreement is phased in, preferential tariffs are lowered sequentially, and positive preferential margins are fully restored. This is visible in Figure 7, which shows that positive preference margins are completely restored after 2018. Thus, we would expect the coefficient β_t to be null or negative for the first years of the implementation of the PTA, and turn positive once

¹Main effects and lower-order interaction terms are absorbed by the fixed effects. In further specifications, we include product-year and country-year fixed effects to account for idiosyncratic trends in imports from specific countries or in specific products. These fixed effects absorb the remaining lower order interactions.

effective preference margins under the agreement are restored, around 2018. To test this expectation, we estimate the following event study specification:

$$\log(Imports_{ict} + 1) = \sum_{\substack{t=2010 \\ t \neq 2011}}^{2022} \beta_t (US_c \times \mathbf{1}\{t = \tau\} \times NegMargin_i) + \alpha_{ic} + \gamma_{it} + \delta_{ct} + \varepsilon_{ict}. \quad (3)$$

where US_c is an indicator equal to one if country c is the United States and zero otherwise, and $NegMargin_i$ is an indicator equal to one for products that experience zero or negative preference margins at any point after the implementation of the US-Colombian PTA. The coefficients β_t trace the differential evolution of U.S. imports for products with negative preference margins in year t , relative to the last pre-PTA year 2011, compared to imports in the same products from non-PTA countries. The specification includes again product-country fixed effects, α_{ic} , as well as product-year fixed effects, γ_{it} , and country-year fixed effects, δ_{ct} . In all specifications, we cluster standard errors at the country-product level.

4.4 Regression Results

Table 3 reports the estimated coefficients from Equation 2. While column (1) includes no controls, columns (2)–(4) progressively introduce product-country, country-year, and product-year fixed effects. The coefficient of interest is the triple interaction, $US_c \times Post_t \times NegMargin_i$, which compares the change in Colombian imports from the U.S. after the PTA's entry into force relative to imports in the same products from countries trading under MFN rates. Crucially, we compare imports in products where preferences under the U.S.-Colombia PTA are eliminated due to MFN reductions relative to imports in products for which a purely positive preference margin materializes. The estimated coefficient on the triple interaction is negative and statistically significant across all specifications, and remains economically meaningful even in the most stringent specification that absorbs product-country, product-year, and country-year variation. In column (4), the coefficient is approximately -0.42. This implies that, following the PTA, U.S. imports of products without an effective tariff preference declined relative to rest-of-world imports by about 34% more than did U.S. imports of products for which a positive preference materialized as negotiated.

This finding is in line with our expectations, as the PTA does not confer any tariff advantage in these products relative to third countries. However, it goes contrary

	log(Imports _{ict} + 1)			
	(1)	(2)	(3)	(4)
$Post_t$	0.131*** (0.036)	-0.098*** (0.037)		
US_c	-0.003 (0.063)			
$NegMargin_i$	0.009 (0.034)			
$US_c \times Post_t$	0.058 (0.079)	-0.288*** (0.102)		
$US_c \times NegMargin_i$	2.424*** (0.075)			
$Post_t \times NegMargin_i$	-0.058 (0.039)	0.112*** (0.040)	0.201*** (0.039)	
$US_c \times Post_t \times NegMargin_i$	-0.867*** (0.091)	-0.901*** (0.109)	-1.051*** (0.110)	-0.422*** (0.117)
Num. Obs.	271 285	271 285	271 285	271 285
Adj. R^2	0.036	0.749	0.758	0.786
Ctry $\times$ Prod FE		✓	✓	✓
Ctry $\times$ Year FE			✓	✓
Prod $\times$ Year FE				✓

Table 3 Trade Effects of Preference Elimination on US imports into Colombia

The table shows the regression results from Equation 2 above. The outcome is the logged import amount of product c from country i in year t . The coefficient of interest is the triple interaction $US_c \times Post_t \times NegMargin_i$, which measures the following triple difference: [US - ROW imports Pre-Post PTA in products that ever have negative margin (for the US)] - [US - ROW imports Pre-Post PTA in products that have a positive margin (for the US)]. Standard errors are clustered at the product-country level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

to existing empirical work on the effect of PTAs on trade, based on negotiated elimination schedules (Besedes, Kohl and Lake, 2020; Baier and Bergstrand, 2007; Dong and Jestrab, 2022). The elimination schedules would suggest that these products are formally covered by the agreement and would typically be expected and treated as receiving preferential treatment. De facto, there is no preferential treatment if one takes into account *unilateral* changes in MFN changes *after* negotiations have been concluded.

Figure 8 depicts the results of the event study specification in Equation 3. In the pre-PTA period, when imports from both the United States and other countries are subject exclusively to MFN tariffs, we find no significant differences between products that subsequently receive a positive preferential margin and those that, despite being formally covered by the U.S.-Colombia PTA, do not confer an effective tariff advantage due to unilateral MFN reductions by Colombia.

Following the entry into force of the PTA, we observe a persistent negative effect

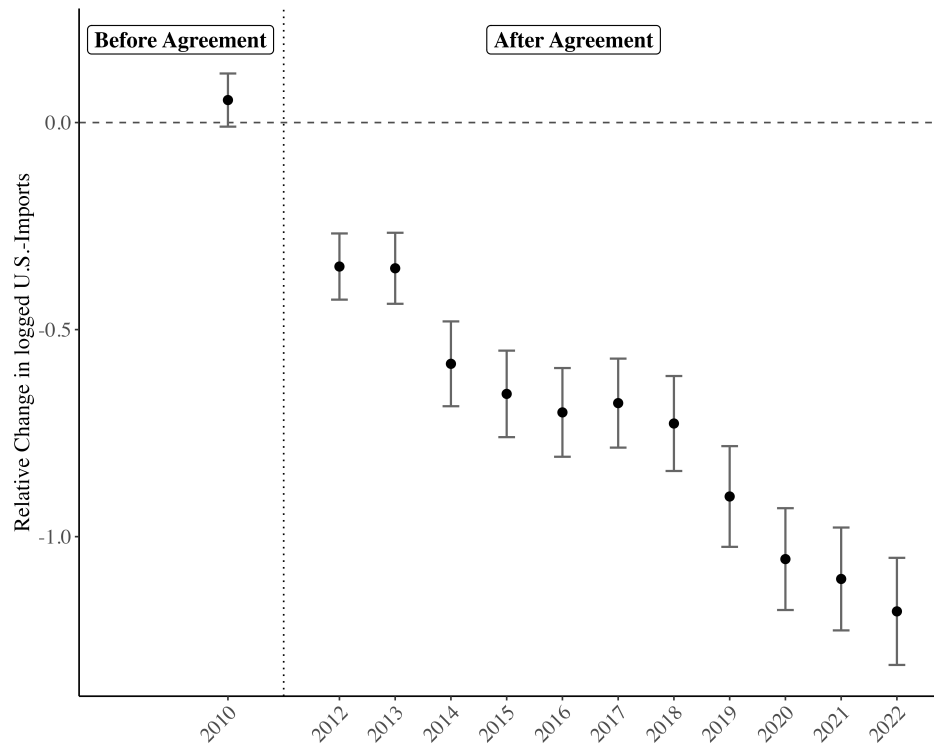


Figure 8
Time-Varying Homogeneous DDD-estimates

The figure plots coefficients from the event study specification in Equation 3, with 2011 as the omitted baseline year. The estimates capture the differential evolution of Colombian imports from the United States relative to imports from countries trading under MFN rates, comparing products that do not receive an effective preferential margin to products in which a positive preferential margin materializes. Whiskers indicate 95% confidence-intervals. Standard errors are clustered at product-country level.

on the import performance of U.S.-originating products without an effective preferential margin. Strikingly, this underperformance persists even after preferential margins are restored, suggesting that the temporary elimination of effective preferences has lasting, negative consequences for intra-PTA trade flows.

4.5 External Validity Across Agreements

In a next step, we test the prevalence of similar MFN changes during the negotiation and implementation of other U.S. PTAs. We replicate the descriptive comparison of negotiated base rates and subsequently applied MFN tariffs leading up to entry-into-force of the agreement for the U.S.–Peru Trade Promotion Agreement (entry-into-force in 2009) and the U.S.–Chile Free Trade Agreement (entry-into-force in 2004). Figure 9 plots both series for the three principal elimination categories.

The contrast is stark. In Peru, approximately 92% of comparable tariff lines retained their negotiated base rate between 2005 and 2007. However, following the 2008 MFN revision, this share fell to 1.5% and remained at that level when the agreement entered into force in 2009. In fact, among the 1,423 tariff lines subject to a 5- or 10-year phase-out, only one still had an MFN rate equal to the base rate specified in the elimination schedule when the agreement entered into force. By contrast, nearly all tariff lines had the same MFN rate as the negotiated base rate when the US–Chile agreement entered into force in 2004.

Thus, while a static base rate exposes an agreement to subsequent MFN liberalization, it does not mechanically eliminate preferential margins. Whether that exposure becomes economically relevant depends on the path of MFN policy between negotiation and implementation. Colombia and Peru illustrate settings in which MFN reductions substantially altered the treatment implied by the negotiated schedule; Chile shows that the same institutional design can coexist with little change in the underlying MFN benchmark. This heterogeneity motivates the broader implications of our findings.

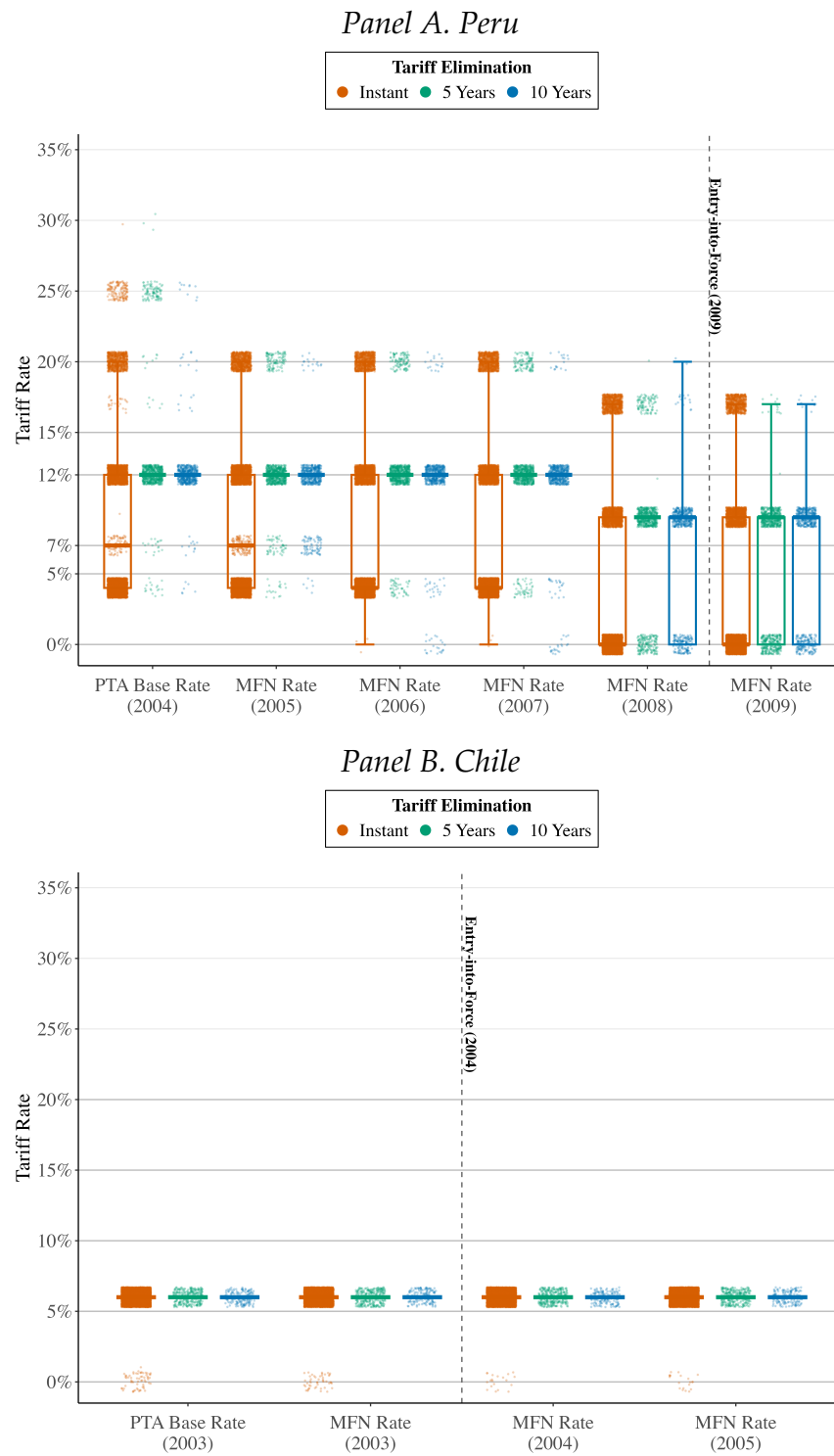


Figure 9
MFN Tariffs Relative to Base Rates

The figure compares the base rates in each agreement's elimination schedule with annual applied MFN rates for the principal elimination categories. The figure illustrates significant heterogeneity in the prevalence of mismatches in between base rates and applied MFN rates. While at entry-into-force of the US-Peru agreement, close to no original base rates were still applicable, the US-Chile agreement saw almost no changes in MFN rates in between negotiation and entry-into-force.

5 Discussion

These results highlight that PTAs do not uniformly confer effective tariff advantages across products during their implementation. In particular, we show that products which are formally covered by a PTA but for which unilateral MFN reductions temporarily eliminate the preferential margin exhibit significantly weaker import performance. This finding is economically intuitive: where no tariff advantage exists, trade should not respond. However, it has important implications for empirical work. Standard approaches typically rely on negotiated tariff schedules and implicitly treat all covered products as effectively receiving preferential treatment. Our results suggest that the principal assumption, that a PTA transfers a discriminatory tariff advantage over third countries, does not necessarily hold (Viner, 1950). As a consequence, including products that are nominally liberalized but do not confer an effective preference introduces a mismeasure of tariff preferences and may substantially downward bias estimates of PTA-induced trade growth. More broadly, the heterogeneous treatment, static vs. updating, of MFN changes during phase-out periods implies that PTAs can generate fundamentally different trajectories of effective tariff advantages. Accounting for these differences is therefore essential when comparing trade effects across agreements.

A similar issue arises for ex-ante evaluations of trade agreements. Quantitative models, such as CGE simulations, typically map negotiated tariff reductions into expected changes in trade (Hertel, 1996). In doing so, they implicitly assume that preferential tariff schedules translate into actual tariff advantages throughout the implementation period. Our findings indicate that this need not be the case. In agreements with static base rates, unilateral MFN reductions can temporarily eliminate effective preferences, implying that simulated tariff reductions overstate the true extent of preferential liberalization. As a result, ex-ante estimates that abstract from this mechanism are likely to be upward biased, particularly in settings where MFN tariffs are adjusted during the implementation phase.

Finally, the distinction between static and updating PTAs has direct policy implications and connects to the broader debate on whether PTAs act as building blocks or stumbling blocks to multilateral trade liberalization (Bhagwati, 1991). In agreements with static base rates, policymakers effectively accept that preferential advantages may erode, or even disappear, if a party to the PTA unilaterally reduces its MFN tariffs. By contrast, PTAs with updating provisions preserve the relative margin of preference, thereby maintaining the intended tariff advantage for intra-PTA products even in the face of multilateral liberalization. Thus, this institutional design feature of PTAs reflects different policy objectives: whether to lock-in a tariff ben-

efit for national export-interests regardless of changes in MFN rates in the partner country, or to accommodate broader multilateral tariff reductions. As such, the interaction between preferential and multilateral liberalization is not only a question of tariff levels, but also of how agreements are designed to respond to tariff changes towards third countries.

6 Conclusion

Preferential trade agreements are generally understood to promote trade among members by lowering tariffs on partner-country imports below the most-favoured-nation (MFN) tariffs faced by third countries. This relative tariff advantage is the standard mechanism through which PTAs are expected to reshape trade. We show that this mechanism can break down during agreement implementation. When preferential tariffs are phased out from a fixed base rate, subsequent unilateral MFN reductions can undercut scheduled preferential rates and temporarily eliminate the negotiated advantage. Whether this occurs depends on a previously overlooked feature of PTA design: whether preferential base rates are updated following MFN reductions or remain fixed. We find that roughly half of the 356 PTAs notified to the WTO do not provide for such updating. In the U.S.-Colombia PTA, this design meant that more than 30% of U.S.-originating imports expected to benefit from the agreement received no effective tariff advantage during its initial phase.

Beyond its empirical relevance, our findings raise important questions about the political economy of PTA design. One possibility is that these design choices reflect the preferences of domestic constituents, such as exporters or import-competing sectors, in line with the political economy literature on trade policy formation (Grossman and Helpman, 1994; Kim, 2017). Alternatively, they may arise from institutional inertia, with legal provisions replicated across agreements without fully accounting for their economic implications (Aischner, Seiermann and Skougarevskiy, 2017; Allee and Elsig, 2019). Distinguishing between these explanations is important, as our results show that seemingly technical differences in legal formulation can have substantial effects on trade outcomes.

The negotiation of PTAs often spans many years and binds significant bureaucratic resources. In addition, firms incur administrative costs to comply with complex rules-of-origin requirements in order to access the tariff benefits enshrined in these agreements (Head, Mayer and Melitz, 2024; Anson et al., 2005). These costs and resources are typically justified by the expectation that PTAs confer meaningful

preferential access. Our results suggest that this expectation is not always met in practice.

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"Preferential" Trade Agreements?

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Supplementary Material

Supplementary Material

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Appendix A: Distribution of MFN Updating Provisions for selected Countries

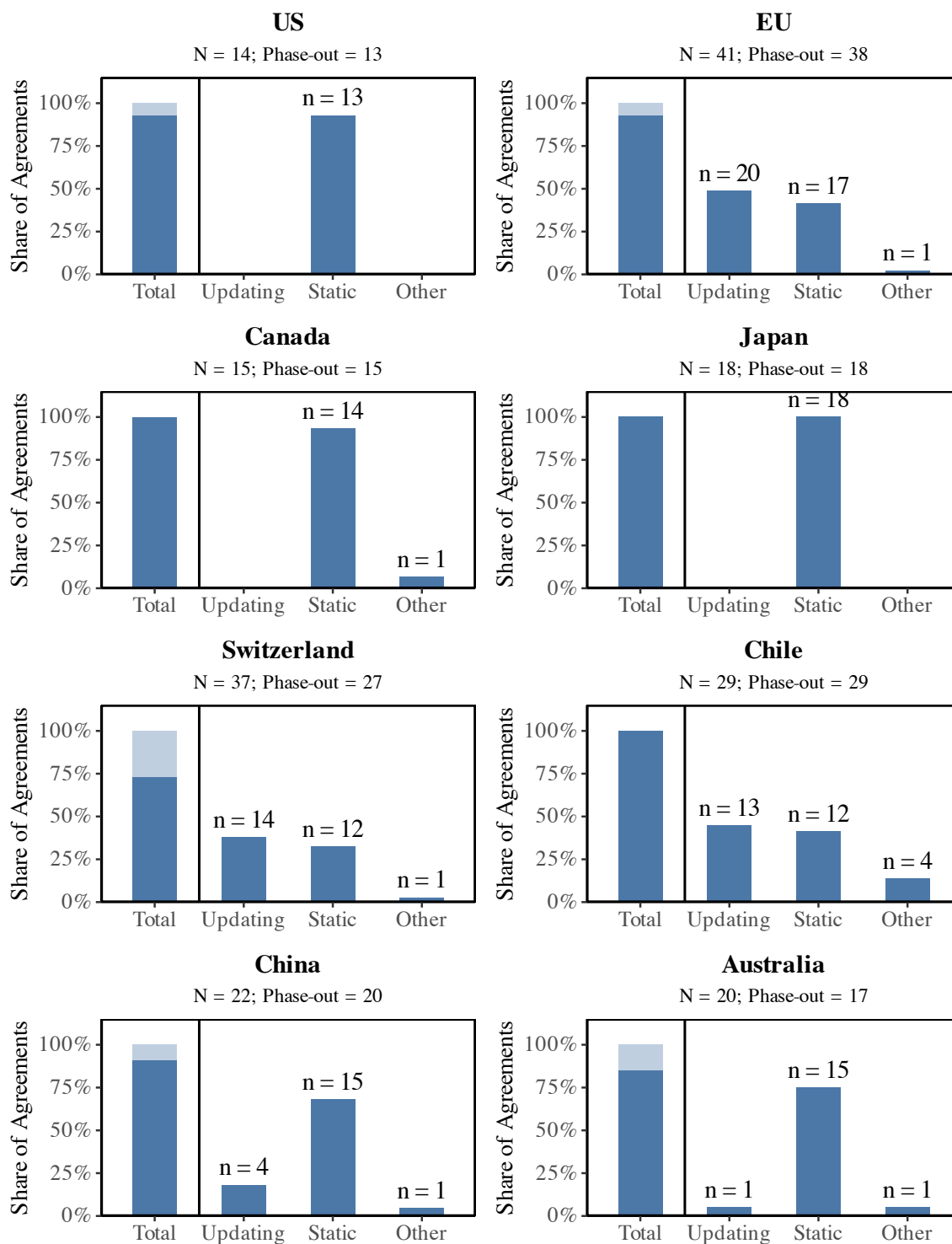


Figure A1 MFN-Updating Provisions Across Selected PTA Members

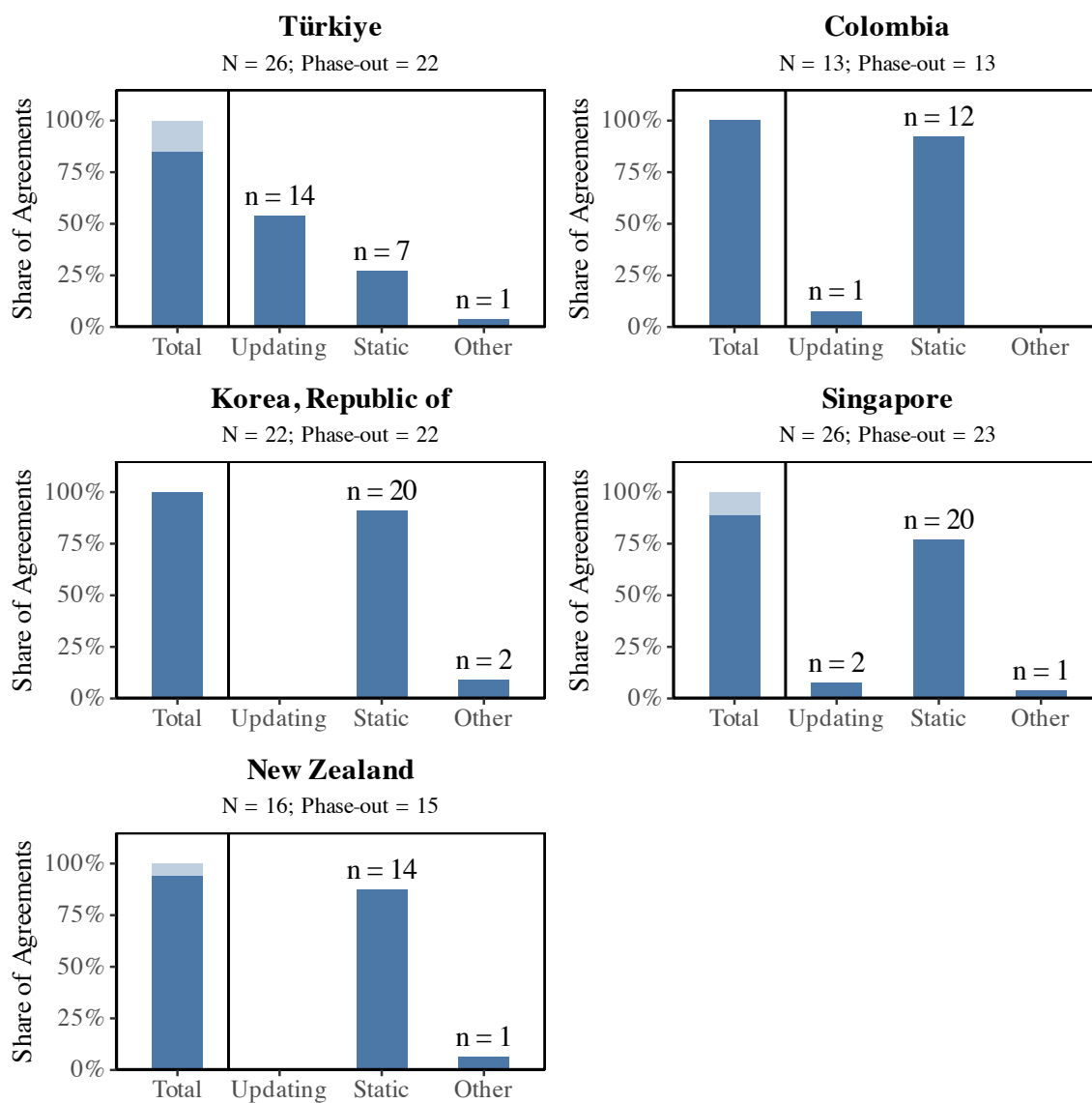


Figure A1 MFN-Updating Provisions Across Selected PTA Members (continued)

Appendix B: Colombia-EU Elimination Schedule

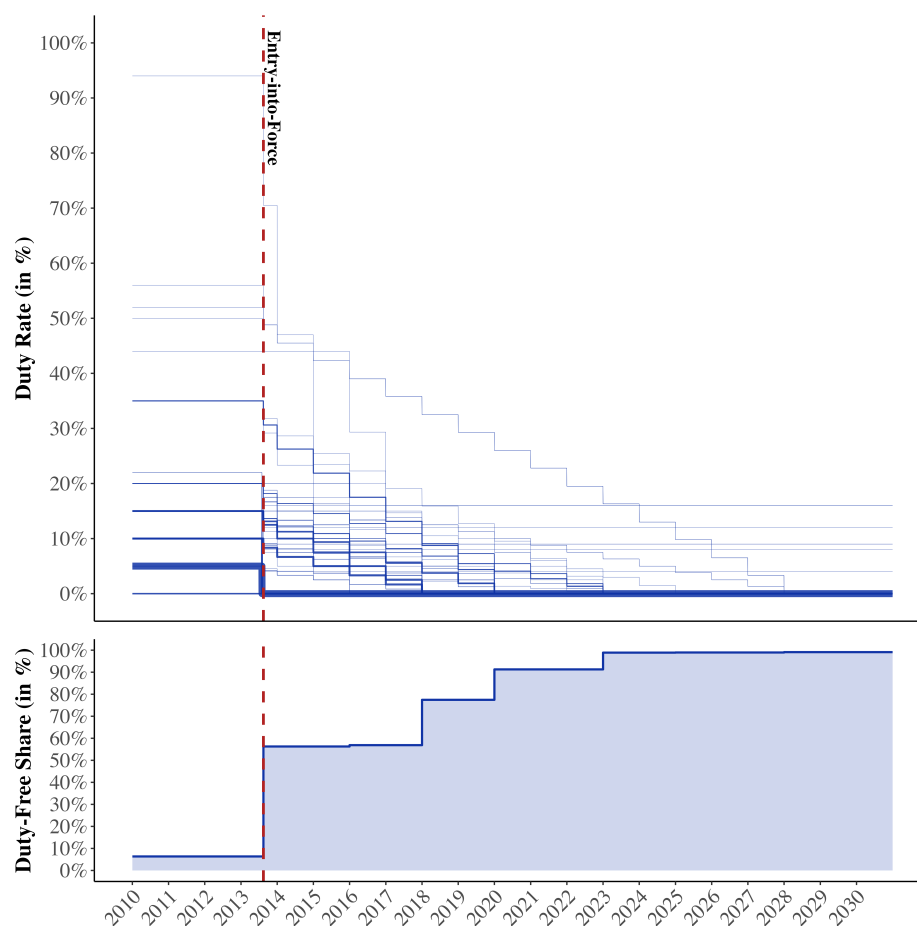


Figure B1

Colombia-EU Elimination Schedule

The figure illustrates the tariff-elimination schedule for EU-originating imports into Colombia as defined in the EU–Colombia Preferential Trade Agreement. Line thickness reflects the pre-PTA trade-weighted share of imports (January 2010–July 2013) across base rate–staging categories. Approximately 55% of pre-PTA imports are subject to immediate tariff elimination upon entry into force, while the remaining 45% are phased out over 6, 8, and 11 years. This latter share of trade is potentially exposed to the elimination of preferential margins through unilateral MFN reductions.

Appendix C: Effective Tariff Preferences of the EU–Colombia PTA

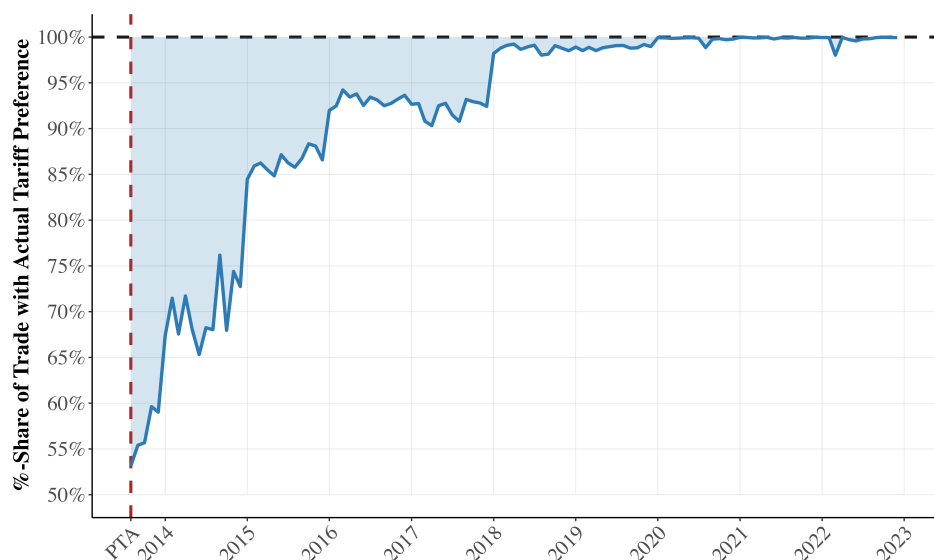


Figure C1

Effective Tariff Preferences under the EU–Colombia PTA

The figure plots the monthly share of EU-originating imports into Colombia that receive a positive preference margin following the entry into force of the EU–Colombia PTA. The share is calculated over imports facing strictly positive MFN tariffs. Due to unilateral MFN reductions by Colombia between negotiation and implementation, preferential rates implied by the agreement exceeded MFN rates for a substantial share of products and trade, eliminating effective tariff advantages. As implementation progressed and preferential rates declined below reduced MFN levels, preference margins were gradually restored.

Appendix D: PTA-Level Correlates of MFN-Updating Provisions

This appendix examines which observable characteristics of preferential trade agreements and the participating countries are associated with the inclusion of MFN-updating provisions. The analysis documents systematic variation in the adoption of these provisions across the composition of negotiating parties, the economic and institutional environment, the pre-agreement tariff regime, and recurring agreement families. We estimate agreement-level linear-probability and logit models with agreement-decade fixed effects. Table D1 summarizes the explanatory variables, their construction and expected associations, and their data sources; the text below describes the common specification and sample.

Table D1 Variables Used in the Agreement-Level Analysis

Variable	Definition	Expected association	Source
North–North / North–South	Indicators for agreement composition; South–South is the omitted category. “North” follows the World Bank’s historical income classification in the year before negotiation, supplemented by a fixed developed-economy classification for earlier agreements. The European Union is treated as one negotiating party.	+ / –	PTA inventory; World Bank
Negotiating complexity	Log number of negotiating parties.	–	PTA membership data
Economic size	Log aggregate GDP of agreement members, averaged over years $t - 3$ through $t - 1$, where t is the negotiation year.	+	World Bank WDI

Continued on next page

Table D1 continued

Variable	Definition	Expected association	Source
Bargaining asymmetry	Largest party's share of aggregate PTA GDP, using the same pre-negotiation GDP window.	+ / –	World Bank WDI
Minimum government effectiveness	Minimum government-effectiveness score among agreement members, averaged over years $t - 3$ through $t - 1$.	+	World Bank WGI
Mean applied MFN tariff	Mean applied MFN tariff over years $t - 5$ through $t - 1$.	+	World Bank WDI / UNCTAD TRAINS
MFN reduction frequency	Frequency with which parties reduced applied MFN tariffs over years $t - 5$ through $t - 1$.	+ / –	World Bank WDI / UNCTAD TRAINS
Economic integration	Intra-PTA bilateral trade divided by aggregate member GDP, averaged over years $t - 3$ through $t - 1$. For agreements negotiated after 2020, the latest available 2018–2020 trade window is used.	+	CEPII Gravity (2022); World Bank WDI

Agreement family	Mutually exclusive indicators for EU/UK, EFTA, Türkiye, Latin American, and Asia-Pacific agreements. “Other” is omitted. Family assignment follows the leading entity in the agreement name.	—	PTA inventory; replication code
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Notes: The unit of observation is a preferential trade agreement. The dependent variable indicates whether the agreement contains an MFN-updating provision. “Expected association” denotes the hypothesized direction of the conditional association with the outcome. GDP measures are averaged over years $t - 3$ through $t - 1$, while tariff measures are averaged over $t - 5$ through $t - 1$. See the text for further details on variable construction and data sources.

Our preferred specifications use the common sample of 252 agreements for which all covariates are observed. All continuous regressors are standardized to have mean zero and standard deviation one in this sample; binary indicators remain coded as zero or one. Party counts are constructed from agreement membership data, while GDP measures are based on contemporaneous membership and aggregate European Union GDP. Higher institutional capacity is expected to facilitate the drafting and administration of updating provisions, whereas greater negotiating complexity may make agreement on contingent tariff adjustments more difficult. Higher tariffs may increase the value of preserving negotiated preferences, while more frequent prior MFN reductions may either increase the demand for protection against preference erosion or make automatic updating commitments more costly. Greater economic integration increases the amount of trade exposed to preference erosion and therefore provides another reason to expect a positive association. The bargaining-asymmetry measure has no unambiguous prediction because a dominant party may either facilitate adoption of a common drafting template or resist provisions that constrain future unilateral tariff policy. Agreement-family indicators capture common drafting traditions; assignment is documented in the replication code.

Every specification includes agreement-decade fixed effects and reports heteroskedasticity consistent HC3 standard errors. Panel A holds the sample fixed at the 252 agreements observed for all preferred-model variables. Panel B applies model-specific listwise deletion and therefore begins with the full set of 273 classified agreements.

Regression Results: Tables D2 and D3 report linear-probability and logit estimates, respectively. Agreements involving northern parties are less likely to contain an MFN-updating provision than South–South agreements, conditional on negotiation decade. The association between tariff levels and the inclusion of an updating provision is positive in the more parsimonious specifications but becomes less precise once agreement-family indicators are included. More broadly, the family coefficients indicate substantial clustering in the adoption of updating provisions across recurring drafting traditions. Given the observational agreement-level design and the limited number of agreements in some families, these estimates are best interpreted as descriptive evidence on where MFN-updating provisions occur rather than as causal effects.

Table D2 Correlates of MFN-Updating Provisions: Linear Probability Models

	Panel A: Common sample			Panel B: Model-specific deletion		
	(1)	(2)	(3)	(1)	(2)	(3)
North–North PTA	-33.31*** (8.21)	-33.79** (15.35)	-34.18** (15.84)	-30.82*** (7.95)	-33.79** (15.35)	-34.18** (15.84)
North–South PTA	-16.24** (7.14)	-15.13* (8.53)	-19.78** (8.45)	-13.71** (6.94)	-15.13* (8.53)	-19.78** (8.45)
Negotiating parties (log, std.)		0.21 (4.28)	-4.06 (4.37)		0.21 (4.28)	-4.06 (4.37)
Total PTA GDP (log, std.)		-5.61 (4.32)	-3.21 (4.14)		-5.61 (4.32)	-3.21 (4.14)
Largest party’s GDP share (std.)		1.48 (3.86)	-1.22 (3.81)		1.48 (3.86)	-1.22 (3.81)
Minimum government effectiveness (std.)		2.82 (5.96)	1.25 (5.79)		2.82 (5.96)	1.25 (5.79)
Mean applied MFN tariff (std.)		4.09 (4.35)	7.33* (4.12)		4.09 (4.35)	7.33* (4.12)
MFN reduction frequency (std.)		-5.44* (3.10)	-3.47 (2.75)		-5.44* (3.10)	-3.47 (2.75)
Intra-PTA trade/GDP (std.)		6.16* (3.57)	-2.89 (4.31)		6.16* (3.57)	-2.89 (4.31)
Asia–Pacific family			-11.36 (7.04)			-11.36 (7.04)
Latin American family			-1.76 (12.50)			-1.76 (12.50)
EU/UK family			43.43*** (12.15)			43.43*** (12.15)
EFTA family			36.55*** (13.09)			36.55*** (13.09)
Türkiye family			35.89** (14.37)			35.89** (14.37)
Observations	252	252	252	273	252	252
R-squared	0.092	0.123	0.299	0.090	0.123	0.299
Agreement-family indicators	No	No	Yes	No	No	Yes
Agreement-decade fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
DV mean (%)	31.7	31.7	31.7	34.1	31.7	31.7

Notes: The dependent variable equals 100 for PTAs with an MFN-updating provision and 0 otherwise, so coefficients are percentage-point associations. Panel A holds the sample fixed at agreements observed for every preferred-model variable. Panel B applies listwise deletion separately by specification, starting from all 273 classified PTAs. South–South PTAs and the Other agreement family are the omitted categories. Continuous regressors are standardized using the common full-model sample. HC3 heteroskedasticity-robust standard errors are in parentheses. The estimates are descriptive, not causal. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D3 Correlates of MFN-Updating Provisions: Logistic Regressions

	Panel A: Common sample			Panel B: Model-specific deletion		
	(1)	(2)	(3)	(1)	(2)	(3)
North–North PTA	-1.91*** (0.61)	-2.04** (0.97)	-3.13** (1.51)	-1.67*** (0.52)	-2.04** (0.97)	-3.13** (1.51)
North–South PTA	-0.71** (0.31)	-0.73* (0.41)	-1.54** (0.65)	-0.59** (0.30)	-0.73* (0.41)	-1.54** (0.65)
Negotiating parties (log, std.)		0.03 (0.22)	-0.30 (0.32)		0.03 (0.22)	-0.30 (0.32)
Total PTA GDP (log, std.)		-0.26 (0.22)	-0.06 (0.28)		-0.26 (0.22)	-0.06 (0.28)
Largest party’s GDP share (std.)		0.09 (0.21)	-0.02 (0.30)		0.09 (0.21)	-0.02 (0.30)
Minimum government effectiveness (std.)		0.16 (0.30)	0.23 (0.46)		0.16 (0.30)	0.23 (0.46)
Mean applied MFN tariff (std.)		0.17 (0.20)	0.45* (0.27)		0.17 (0.20)	0.45* (0.27)
MFN reduction frequency (std.)		-0.28* (0.16)	-0.19 (0.18)		-0.28* (0.16)	-0.19 (0.18)
Intra-PTA trade/GDP (std.)		0.30 (0.19)	-0.29 (0.26)		0.30 (0.19)	-0.29 (0.26)
Asia–Pacific family			-1.47** (0.74)			-1.47** (0.74)
Latin American family			-0.13 (0.75)			-0.13 (0.75)
EU/UK family			2.92*** (0.92)			2.92*** (0.92)
EFTA family			2.36** (0.98)			2.36** (0.98)
Türkiye family			1.70* (0.91)			1.70* (0.91)
Observations	252	252	252	273	252	252
Agreement-family indicators	No	No	Yes	No	No	Yes
Agreement-decade fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
McFadden pseudo-R-squared	0.079	0.104	0.290	0.075	0.104	0.290

Notes: Entries are log-odds coefficients from logistic regressions; the dependent variable equals one for PTAs with an MFN-updating provision and zero otherwise. Panel A holds the sample fixed at agreements observed for every preferred-model variable. Panel B applies listwise deletion separately by specification, starting from all 273 classified PTAs. South–South PTAs and the Other agreement family are omitted. Continuous regressors are standardized using the common full-model sample. HC3 heteroskedasticity-robust standard errors are in parentheses. The estimates are descriptive, not causal. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

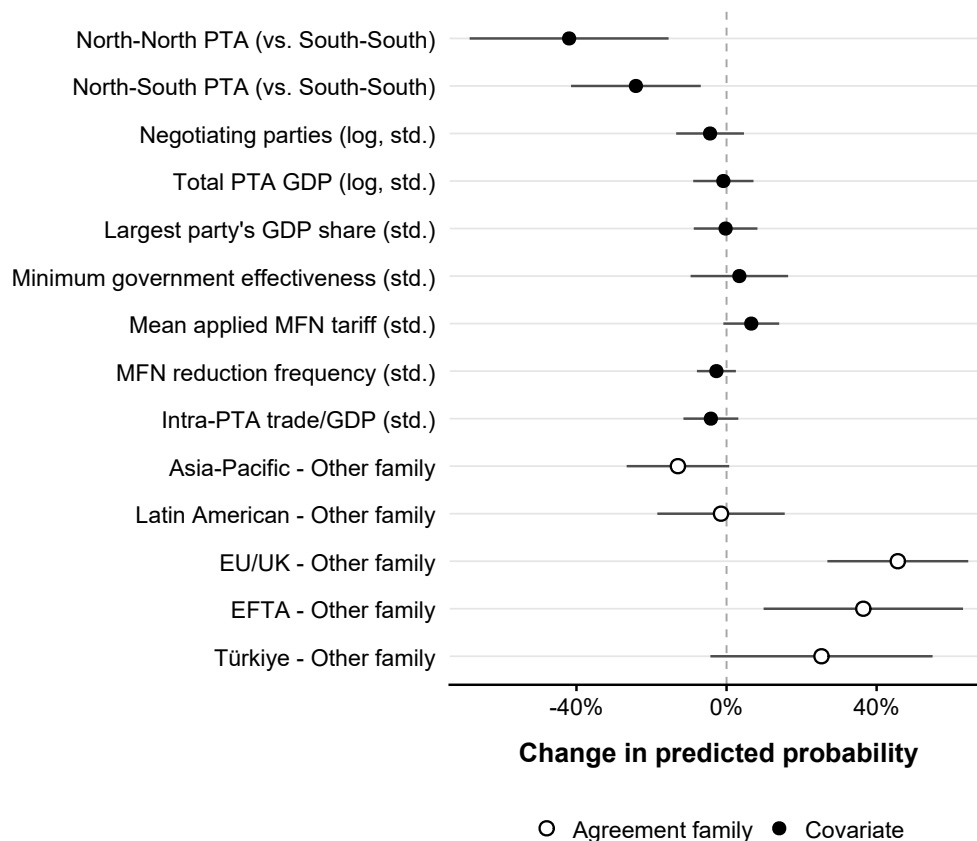


Figure D1

Agreement-Level Correlates of MFN-Updating Provisions

The figure reports average marginal effects from Model 3 in Table D3, estimated on the 252 PTAs for which complete data on all covariates are available. Points indicate estimated changes in the probability that an agreement contains an MFN-updating provision; horizontal lines show 95% confidence intervals based on HC3 heteroskedasticity-robust standard errors. Effects are expressed in percentage points. Continuous covariates are standardized, so their estimates represent a one-standard-deviation increase. Effects for North-North and North-South PTAs are relative to South-South PTAs; agreement-family effects are relative to the “Other” family (United States, Canada, Ukraine, Eurasian Economic Union, AfCFTA, UAE, Switzerland, Iceland, Guatemala, GCC, El Salvador, Egypt, Brazil, Jordan, Viet Nam, Mauritius, Mozambique, CEFTA, CPTPP, PACER Plus, Pacific Alliance, PICTA, Agadir Agreement, and Trans-Pacific Partnership). The model includes agreement-decade fixed effects, which are not displayed.

Appendix E: Mapping Colombian MFN Tariffs to the Elimination Schedule

E.1 The harmonization problem

he creation of Figure 5 requires the mapping of an MFN tariff for the same product at several points in time. The natural unit of observation is the tariff line in the Colombia–United States preferential trade agreement (PTA) elimination schedule. That schedule records the negotiated base rate and phase-out category using Colombia’s national tariff nomenclature of 2004. We therefore keep the original schedule code fixed throughout the analysis. For each tariff decree, the task is to map the tariff lines and MFN rates reported under the nomenclature then in force back to this fixed 2004 product identifier.

This mapping is not a conventional merge with an imperfectly formatted key. National tariff nomenclatures change as products are divided, combined, or reassigned across revisions. Consequently, a code in the 2004 schedule can correspond to one code, several codes, or no immediately observable code in a later decree. A single concordance step may be insufficient when the decree and the elimination schedule are separated by several nomenclature revisions. These difficulties become more important as the decree year moves farther from the 2004 schedule: the share of schedule lines that can be joined directly falls from 96.03 percent for the 2004 decree to 77.54 percent for the 2016 decree (Table E1).

Two additional features of the elimination schedule require special treatment. Industrial products and some agricultural products are reported at ten digits, whereas other agricultural commitments are reported at eight digits. Some agricultural lines also carry nonnumeric suffixes that distinguish commitments within the same eight-digit heading. We preserve the original schedule entry, including any suffix, as the product identifier. For matching only, an agricultural suffix is removed and the corresponding eight-digit prefix is used. Thus, the procedure never replaces the original elimination-schedule code with a MFN-decree code; decree codes are retained separately as evidence for the match.

E.2 Ordered matching procedure

e apply the same procedure separately to Decree 4341 of 2004, Decree 4589 of 2006, Decree 4927 of 2011, and Decree 2153 of 2016.

The algorithm proceeds in four stages. First, ten-digit schedule lines are joined exactly to ten-digit decree lines. Eight-digit agricultural lines, including suffix-bearing

entries after removal of the suffix, are joined to the first eight digits of decree lines. Second, schedule lines not resolved directly are passed through the Colombian nomenclature concordance once. A translated ten-digit code is compared with the exact decree code; for an eight-digit agricultural schedule line, the translated eight-digit prefix is also considered. Third, remaining lines are followed iteratively through the concordance network for up to ten steps. Links are allowed in both recorded directions because the objective is to connect two nomenclature vintages rather than to impose an assumed chronological direction on an individual concordance record. Cycles are prohibited within a search path. A path stops once it reaches a valid decree endpoint, and the full path and number of translation steps are retained for audit purposes. Fourth, cases that remain unresolved or ambiguous are reviewed manually as described below.

At every automatic stage, the procedure collects all decree lines reachable from the original schedule line. A correspondence is assigned automatically only when all reachable decree lines imply the same MFN rate. This includes both one-to-one matches and one-to-many code mappings whose descendant lines all carry an identical rate. When reachable decree lines carry different rates, the original schedule line is classified as ambiguous and sent to manual review. Lines for which the concordance search produces no decree endpoint are also sent to manual review. Importantly, the hierarchy is ordered: once a code yields an ambiguous set of candidates at a given stage, it is reviewed rather than allowed to select a potentially more convenient result from a later stage.

E.3 Manual resolution and conservative rate assignment

Manual review is necessary when a concordance identifies feasible code relationships but cannot determine which descendant rate should represent an earlier, broader product category when that category has subsequently been split. We apply a common, pre-specified rule across decree years. Suppose that several tariff lines in a newer nomenclature map back to one original 2004 elimination-schedule code and that these lines carry different MFN rates.

First, if exactly one of the candidate MFN rates equals the negotiated base rate recorded for the original schedule line, we assign the corresponding decree code and MFN rate to that line. Second, if all candidate MFN rates are below the original base rate, we assign the highest available candidate MFN rate. This is the most conservative feasible assignment for tests of MFN liberalization: it minimizes the measured decline from the base rate and therefore avoids mechanically overstating the extent to which the applied MFN tariff fell relative to the negotiated benchmark.

The same logic is used when manual resolution is needed after a longer concordance path. Each judgment is stored in a persistent decision file containing the original elimination code, the selected decree code where one can be identified, the assigned MFN rate, its relationship to the base rate, and a review note. Completed manual decisions override automatic results, which makes the resolution explicit and reproducible rather than dependent on row order or an implicit many-to-many join.

The resulting rate is then classified relative to the original base rate as equal to, below, or above that benchmark. This classification is a comparison made after the correspondence has been established; it is not, except for the stated manual tie-break, used to manufacture an automatic match. The procedure produces one corresponding MFN rate for each of the 6,922 original elimination-schedule lines in every decree year.

Table E1 Mapping the Tariff Decrees to the 2004 Elimination-Schedule Nomenclature

Tariff decree	Direct match	One-step translation	Multi-step translation	Manual decision	Total matched
Decree 4341 (2004)	6,647 (96.03%)	193 (2.79%)	2 (0.03%)	80 (1.16%)	6,922 (100%)
Decree 4589 (2006)	5,799 (83.78%)	888 (12.83%)	21 (0.30%)	214 (3.09%)	6,922 (100%)
Decree 4927 (2011)	5,559 (80.31%)	999 (14.43%)	53 (0.77%)	311 (4.49%)	6,922 (100%)
Decree 2153 (2016)	5,367 (77.54%)	1,075 (15.53%)	91 (1.31%)	389 (5.62%)	6,922 (100%)

Notes: The unit is an original tariff line in the 2004 Colombia–United States PTA elimination schedule. Cells report the number of schedule lines, with the share of all 6,922 lines in parentheses. “Direct” combines exact ten-digit matches, ordinary eight-digit agricultural matches, and suffix-bearing agricultural matches made through their eight-digit numeric prefix. “One-step” and “multi-step” use the Colombian nomenclature concordance. “Manual” includes ambiguous mappings with heterogeneous candidate rates and cases without an automatic decree endpoint. Shares may not sum to exactly 100 because of rounding. Source: authors’ calculations from the decree-specific match-method summaries.

Table E1 provides a diagnostic for the underlying harmonization problem. Translation and manual review become progressively more important as the nomenclature in the tariff decree moves farther from the 2004 schedule. Nevertheless, the increasing reliance on concordances does not itself determine the assigned rate: translated mappings remain automatic only when all feasible decree endpoints agree on the MFN rate. Differences in MFN rates, rather than the mere existence of multiple

descendant codes, is what triggers manual judgment.